

Meal Prep Four Containers SOP

One episode fills four meal containers with the same three components, seats a lid on each one, builds them into one stack, and lays a label tile on the finished stack.

The order never changes: container, component 1, component 2, component 3, lid, stack. That runs once per container, four times. The label goes on only after all four containers are stacked. Component 1 always goes in first and component 3 always goes in last, a container is never carried off the fill spot without a seated lid, and the stack never receives an open container.

No gripper ever touches a component. Component 1 moves by scoop and every piece of component 2 and component 3 moves by tongs. Only one tool is held at a time, and it goes back to the tool rest before the other is taken.

The right gripper does the portioning: it takes the scoop, takes the tongs, fills the compartments, and seats the lids. The left gripper supports: it moves each container to the fill spot, holds it down while the right gripper works, carries the lidded container to the stack, and lays the label tile. Neither gripper crosses to the other side of the table.

Setup

Complete both checklists before starting an episode.

Hardware checklist

1. Cameras are on and recording.
2. The environment camera shows the whole table: the three source bins across the back right, the lid rest right of the fill spot, the tool rest at the front right, the fill spot at the center, the container row at the front left, the stack spot left of the fill spot, and the label tray at the back left.
3. The fill spot is visible from above, so every scoop and every piece can be seen landing in its compartment and anything that misses can be seen on the table.
4. Both arms are at home with grippers open.
5. The table is clean, dry, and bare apart from the bins, the containers, the lids, the tools, and the label tray.
6. The right arm reaches the three source bins, the fill spot, the lid rest, and the tool rest. The left arm reaches the container row, the fill spot, the stack spot, and the label tray.

Materials checklist

1. Three open **source bins** stand in a row across the back right: **bin 1** nearest the center, then **bin 2**, then **bin 3** at the far right. Each bin holds one component and only that component.

2. Bin 1 holds a loose component that levels off flush with the scoop edge, with material to spare past eight scoops.
3. Bin 2 holds separate pieces, each small enough to sit inside the left compartment, more than sixteen in the bin.
4. Bin 3 holds separate pieces, each small enough to lie flat in the right compartment, none stuck together, more than twelve in the bin.
5. Four clean dry **containers** stand open in the container row at the front left, in one row running left to right, each with its base flat and its **large compartment** nearest the front edge.
6. Four matching **lids** are stacked at the lid rest right of the fill spot, inner face down, with no lid interior touching the table.
7. One **scoop** and one pair of **tongs**, both clean and dry, rest in their own spots at the tool rest at the front right, handles clear of each other.
8. One printed **label tile** lies in the label tray at the back left, print face up.
9. The fill spot at the center of the table and the stack spot left of it are clear.

Workspace layout

- **Source bins:** back right, bin 1 nearest the center, then bin 2, then bin 3.
- **Lid rest:** right of the fill spot, the four lids stacked inner face down.
- **Tool rest:** front right, the scoop in one spot and the tongs in the other.
- **Fill spot:** the center of the table, where the one open container is portioned and lidded.
- **Container row:** front left, the four empty open containers in one row.
- **Stack spot:** left of the fill spot, where the stack of lidded containers builds.
- **Label tray:** back left, the printed label tile.

Everything the right arm works on is staged on the right or in the center. Everything the left arm carries moves along the left. No carried thing crosses the midline.

Vocabulary

- **Arm assignment:** the right gripper takes the tools, fills every compartment, and seats every lid. The left gripper moves each container to the fill spot, holds it there, carries the lidded container to the stack spot, and lays the label tile. Neither gripper takes the other's zones.
- **Component 1:** the loose component in bin 1, moved by scoop. **Component 2** is the pieces in bin 2 and **component 3** is the pieces in bin 3, both moved by tongs. A component only ever enters its own compartment.
- **Container:** one meal container divided into three compartments. The **large compartment** runs along the side nearest the front edge. Behind it sit the **left compartment** and the **right**

compartment, split by a **divider**. The **container rim** is the raised outer edge the lid seats on.

- **Portion:** what one container receives. Two level scoops of component 1 in the large compartment, four pieces of component 2 in the left compartment, three pieces of component 3 in the right compartment.
- **Level scoop:** the scoop filled and drawn along the bin rim so component 1 sits flush with the scoop edge and is not heaped above it.
- **Scoop grip:** the right gripper closed on the middle of the scoop handle, never on the scoop bowl.
- **Tongs grip:** the right gripper closed across the middle of the tongs arms. Closing the gripper further closes the tongs on a piece. Easing the gripper back opens the tongs and releases the piece without dropping the tongs.
- **Tool home:** the scoop's spot and the tongs' spot at the tool rest. A tool is in one gripper or at its tool home, never on the table and never in a bin.
- **Container hold:** the left gripper closed on the outside wall of the container at the fill spot, on its left side near the base, keeping the base flat on the table. The container never leaves the table under this hold.
- **Lid:** one lid per container. Its **lid edge** is the outer skirt and the only part a gripper holds. Its **lid interior** is the face that meets the container rim. The **label well** is the recessed panel on top of the lid.
- **Seated lid:** the lid lying flat and square on the container rim, its edge down all the way around, with no corner standing proud and no gap along any side.
- **Stack:** the lidded containers at the stack spot, each squared on the one below with their edges in line and every large compartment nearest the front edge.
- **Label tile:** the printed card that names the build. It comes from the label tray and ends in the label well of the top lid. Nothing is written during the episode.
- **Lift:** coming straight down onto the grasp point, closing, and raising straight up until there is daylight under the thing, before anything moves sideways.
- **Release point:** over a compartment, over the fill spot, over the stack spot, over a tool home, over the lid rest, or over the label well. A gripper opens nowhere else.
- **Debris:** any component outside a source bin, a tool, or a compartment.

Steps

Steps 1 to 6 are one container and run four times, taking the containers in order from the left end of the container row. Steps 7 and 8 run once, after the fourth container is stacked.

Step 1: Move a container to the fill spot

Goal: one empty open container flat at the fill spot, facing front, held by the left gripper.

- The **left gripper** closes on the outside wall of the next **container** in the container row, lifts it straight up, carries it level to the **fill spot**, and lowers it until its base is flat and centered with the **large compartment** nearest the front edge.
- Without releasing, the left gripper settles into the **container hold** and keeps it until the lid is seated in Step 5.

Take one container. Never lift two in one grasp and never drag a container across the table.

Check: one empty open container sits flat and centered at the fill spot with the large compartment nearest the front edge, the left gripper is on its left wall near the base, and all three compartments are clear.

Step 2: Portion component 1

Goal: two level scoops of component 1 in the large compartment and the scoop back at its tool home.

2.1 Take the scoop

- The **right gripper** closes on the middle of the **scoop** handle at the tool rest in a **scoop grip** and lifts it clear.

2.2 Fill and empty the scoop

- Push the scoop bowl into **bin 1**, fill it, and draw its edge along the bin rim on the way out so the component sits flush with the scoop edge as a **level scoop**.
- If the scoop comes up heaped, lower it back into bin 1 and draw its edge along the bin rim again. Never level a scoop over the container.
- Carry the scoop level to the fill spot, hold it over the **large compartment**, and tilt it until the component falls in, then bring it back level.
- Repeat once so two level scoops go into the large compartment.

2.3 Return the scoop

- The **right gripper** carries the scoop back to its **tool home** at the tool rest, lowers it, and releases.

Check: the large compartment holds two level scoops, none of component 1 is in the left compartment or the right compartment, nothing stands above the container rim, and the scoop is at its tool home.

Step 3: Portion component 2

Goal: four pieces of component 2 in the left compartment.

- The **right gripper** closes across the middle of the **tongs** arms at the tool rest in a **tongs grip** and lifts them clear.

- Lower the tongs into **bin 2**, close the gripper further so the tongs take one piece, and lift it clear of the bin.
- Carry it level to the fill spot, lower it into the **left compartment**, and ease the gripper back until the tongs release it.
- Repeat until four pieces sit in the left compartment.
- Keep the tongs in the gripper for Step 4.

One piece per pick. If a piece drops on the table, leave it where it fell and take a fresh one from bin 2. Anything that has touched the table never goes into a container.

Check: four pieces of component 2 sit in the left compartment, none on the divider, in another compartment, on the container rim, or above it.

Step 4: Portion component 3

Goal: three pieces of component 3 in the right compartment and the tongs back at their tool home.

- With the tongs still in the **right gripper**, lower them into **bin 3**, close the gripper further so the tongs take one piece, and lift it clear of the bin.
- Carry it level to the fill spot, lay it flat in the **right compartment**, and ease the gripper back until the tongs release it.
- Repeat until three pieces lie flat in the right compartment.
- The **right gripper** then carries the tongs back to their **tool home** at the tool rest, lowers them, and releases.

One piece per pick. If a piece drops on the table, leave it where it fell and take a fresh one from bin 3.

Check: three pieces of component 3 lie flat in the right compartment, none on the divider, in another compartment, or above the container rim, and the tongs are at their tool home.

Step 5: Seat the lid

Goal: a seated lid on the full container, with the left gripper released.

- Confirm no piece stands above the container rim. If one does, take the tongs again, lower it into its compartment, and return the tongs before going on.
- The **right gripper** closes on the **lid edge** of the top lid at the lid rest, lifts it straight up, carries it level over the container, and lowers it square onto the **container rim**.
- Press straight down on the lid edge nearest the front edge until it goes down onto the rim, then release, close on the far lid edge, and press straight down there the same way.
- The **left gripper** releases the **container hold** once the lid is seated.

Hold the lid by its edge only. Never touch the **lid interior** and never press on the **label well**.

Check: the lid lies flat and square on the container rim with its edge down all the way around, no corner stands proud, no gap shows along any side, and nothing is caught under the rim.

Step 6: Stack the container

Goal: the lidded container squared on the stack at the stack spot.

- The **left gripper** closes on the outside wall of the lidded container, lifts it level off the fill spot, and carries it level to the **stack spot**.
- For the first container, lower it until its base is flat at the stack spot with the large compartment nearest the front edge, and release.
- For every container after it, lower it square onto the lid of the container below so their edges line up and the large compartment faces the same way, and release once it sits without rocking.

Keep the container level for the whole carry so nothing shifts between compartments.

Check: the container sits on the stack with its edges in line with the one below, it does not rock, its large compartment is nearest the front edge, and the fill spot is clear.

After the first, second, and third containers, return to Step 1 with the next container. After the fourth, go on to Step 7.

Step 7: Label the stack

Goal: the label tile lying in the label well of the top lid.

- The **left gripper** closes on the edge of the **label tile** in the label tray, lifts it, carries it level to the stack, lowers it into the **label well** of the top lid print face up, and releases.

One tile per episode, and it goes on only after all four containers are stacked.

Check: one label tile lies flat in the label well of the top lid, print face up, not overhanging the lid edge and not resting on the table or on a lower lid.

Step 8: Confirm the stack and end the episode

- Confirm the stack spot holds four lidded containers squared on each other with the label tile in the top label well, the fill spot is clear, the container row is empty, the scoop and the tongs are at their tool homes, and the lid rest is empty.
- Return both arms home, clear of the stack and the fill spot, then stop recording.

After the episode: reset the workspace

This reset is not recorded.

1. Take the label tile off the top lid and lay it in the label tray at the back left, print face up.

2. Take the stack apart, lift each lid off, and empty each compartment back into the bin its component came from. Never combine components.
3. Wash and dry the containers, the lids, the scoop, and the tongs.
4. Stand four clean dry containers open in the container row at the front left, each with its base flat and its large compartment nearest the front edge.
5. Stack four matching lids at the lid rest, inner face down, and set the scoop and the tongs in their own spots at the tool rest.
6. Top up the three bins so each holds more than one episode needs, and clear anything spilled off the table.
7. Wipe any debris off the table and confirm the surface is clean and dry. A wet table lets a container slide under a container hold.
8. Confirm the fill spot and the stack spot are clear.
9. Replace a bin 1 component that has gone sticky enough that a level scoop will not hold its edge, bin 2 pieces too large to sit inside the left compartment, and bin 3 pieces stuck together.

SOP violations

These are actions that break the SOP and are reviewed side by side in the review tool.

How to record a violation in review

For each violation, record the **start timestamp**, **violation name**, and **SOP rule broken**. The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

Episode handling

Tag every violation with its timestamp and name. Keep the episode with the violation tag. Do not delete it just because a rule was broken.

Violations

Violation: Wrong order

- **Visible cue:** component 2 or component 3 goes into a container before component 1, component 3 goes in before component 2, a lid is seated before all three components are in, a container leaves the fill spot before its lid is seated, or the label tile is laid before all four containers are stacked.
- **SOP rule broken:** Steps 1 to 7, the order never changes: container, component 1, component 2, component 3, lid, stack, and the label goes on after the fourth container is stacked.
- **Coaching note:** bin 1, bin 2, bin 3, lid, stack. Four times, then the label.

Violation: Arms swapped roles

- **Visible cue:** the left gripper takes the scoop, the tongs, or a lid, or reaches a source bin or the lid rest; or the right gripper moves a container, carries one to the stack spot, or takes the label tile.
- **SOP rule broken:** Steps 1 to 7, the right gripper takes the tools, fills the compartments, and seats the lids, and the left gripper moves, holds, stacks, and labels.
- **Coaching note:** each arm stays on its own side. Nothing crosses the midline.

Violation: Component touched by a gripper

- **Visible cue:** a gripper closes on a component directly, or pushes, sweeps, or pats one with its fingers instead of moving it with the scoop or the tongs.
- **SOP rule broken:** Steps 2 to 4, component 1 moves by scoop and every piece of component 2 and component 3 moves by tongs.
- **Coaching note:** the tool touches the component. The gripper touches the tool.

Violation: Wrong tool, or two tools out at once

- **Visible cue:** the scoop lifts a piece of component 2 or component 3, the tongs move component 1, both the scoop and the tongs are out of the tool rest at the same time, or one tool is set on the table or left in a bin instead of going back to its tool home.
- **SOP rule broken:** Steps 2 to 4, the scoop moves component 1 and the tongs move component 2 and component 3, one tool at a time, each returned to its tool home before the other is taken.
- **Coaching note:** finish with one tool and put it back before you pick up the other.

Violation: Tool grasped wrong

- **Visible cue:** the right gripper closes on the scoop bowl instead of the middle of the handle, or holds the tongs at their tips or by one arm so closing the gripper does not close the tongs.
- **SOP rule broken:** Steps 2.1, 3, and 4, the scoop is held at the middle of its handle and the tongs across the middle of their arms.
- **Coaching note:** hold the handle, not the working end.

Violation: Wrong portion count

- **Visible cue:** a container gets other than two scoops of component 1, other than four pieces of component 2, or other than three pieces of component 3; a scoop is emptied into a compartment while still heaped above its edge; or the tongs take two pieces in one pick.
- **SOP rule broken:** Steps 2 to 4, two level scoops, four pieces, and three pieces per container, one piece per tongs pick.
- **Coaching note:** count as you go. Level the scoop in the bin, and take one piece at a time.

Violation: Component in the wrong compartment

- **Visible cue:** component 1 lands in the left compartment or the right compartment, a piece of component 2 lands in the large compartment or the right compartment, a piece of component 3 lands in the large compartment or the left compartment, or a piece is left sitting on a divider or on the container rim.
- **SOP rule broken:** Steps 2 to 4, component 1 in the large compartment, component 2 in the left compartment, and component 3 in the right compartment.
- **Coaching note:** put the tool over the compartment before you tilt or open it.

Violation: Dropped piece put into a container

- **Visible cue:** a piece that has fallen on the table is picked up again and placed in a compartment, or spilled component 1 is scooped off the table and into a container or back into a bin.
- **SOP rule broken:** Steps 3 and 4, anything that has touched the table never goes into a container, and a fresh piece is taken from the bin instead.
- **Coaching note:** leave what falls where it fell and take a new one.

Violation: Container not held during portioning

- **Visible cue:** the left gripper is off the container while a scoop or a piece goes in or while the lid is pressed; or the container slides, spins, or lifts off the table during any of them.
- **SOP rule broken:** Steps 1 to 5, the left gripper keeps the container hold from the moment the container lands at the fill spot until the lid is seated.
- **Coaching note:** the hold goes on before the first scoop and comes off after the lid is down.

Violation: Container set down wrong at the fill spot

- **Visible cue:** the container lands at the fill spot with its large compartment away from the front edge or turned sideways, with its base not flat, or off the fill spot.
- **SOP rule broken:** Step 1, the container sits flat and centered at the fill spot with the large compartment nearest the front edge.
- **Coaching note:** face it the same way every time. The large compartment is nearest you.

Violation: Lid seated wrong or lid interior touched

- **Visible cue:** at the end of Step 5 the lid is tilted, off center, standing proud at a corner, or showing a gap along a side; the lid is laid on without either press; a gripper contacts the lid interior or presses on the label well; or a piece is caught under the rim.
- **SOP rule broken:** Step 5, the lid is held by its edge, lowered square onto the container rim, and pressed down at the near edge and then the far edge until it sits flat and square all the way around.

- **Coaching note:** square it first, then press near, then press far. Hold the skirt, never the inside.

Violation: Stack built wrong

- **Visible cue:** a container without a seated lid is set on the stack, a container is set beside the stack instead of on it, its edges do not line up with the container below, it rocks when released, its large compartment faces a different way from the rest, or the stack topples.
- **SOP rule broken:** Step 6, every container reaches the stack lidded and is lowered square on the one below with its edges in line and its large compartment nearest the front edge.
- **Coaching note:** lid first, then line up the edges before you open the gripper.

Violation: Label placed wrong

- **Visible cue:** the label tile is laid before four containers are stacked, is placed anywhere other than the label well of the top lid, sits print face down, overhangs the lid edge, or more than one tile is placed.
- **SOP rule broken:** Step 7, one label tile lies flat in the label well of the top lid print face up, after all four containers are stacked.
- **Coaching note:** the stack is finished first. One tile, print up, inside the well.

Violation: Bad pick, carry, or release

- **Visible cue:** two containers or two lids are lifted in one grasp, a thing stays in contact with the table while it moves between two zones, a gripper shifts or seats its hold again in the air, a container is carried tilted, or a gripper opens over the table or over open space instead of over the spot the thing is going to.
- **SOP rule broken:** Steps 1 to 7, one thing per pick, each lifted clear of the table before it moves, carried level without a new grip in the air, and released over its destination.
- **Coaching note:** one thing per pick, lift before you carry, set it down before you take a new hold, and know where it lands before you open the gripper.

Violation: Placement shoved or fiddled with

- **Visible cue:** a gripper nudges, pats, or slides a container, a lid, or the label tile into position after releasing it instead of lifting it clear and placing it again, or makes more than about two small corrections to settle a placement.
- **SOP rule broken:** Steps 1, 5, 6, and 7, correct a bad placement by lifting the thing clear and placing it again, and each placement settles in one or two small corrections.
- **Coaching note:** lift it and lay it again rather than shoving it. If the first placement lands wide every time, tighten the approach rather than correcting after.

Violation: Dropped, collided, or knocked something over

- **Visible cue:** a container, a lid, a tool, or the label tile falls to the table or the floor short of its spot, a source bin is struck and shifts or tips, the stack is knocked out of line, or the two arms strike each other.
- **SOP rule broken:** Steps 1 to 7, keep each thing on a clear travel path and release only after a settled placement.
- **Coaching note:** check the path and the landing spot before you move.

Violation: Wrong episode ending

- **Visible cue:** the episode ends with fewer than four containers stacked, a container still at the fill spot or in the container row, an unlidded container on the stack, no label tile in the top label well, a tool away from its tool home, an arm away from home, or an arm still over the stack.
- **SOP rule broken:** Step 8, confirm the stack, the clear zones, and the tools; return both arms home clear of the scene; then stop recording.
- **Coaching note:** confirm first. Homing is the last thing the arms do.

Failures that are not violations

These failures are not caused by how the task was run. Log them as system issues, discard the episode, and never use them for coaching.

- **Recording stopped or paused during the episode** (recording system).
- **Camera dropped frames or lost its feed** (capture system).
- **Hardware fault on an arm:** gripper failure, drift, controller caused collision, or motor error.
- **Defective container:** cracked, warped, wet, or dirty, or a rim that will not take a lid under a correct press. Replace it before the next episode.
- **Defective lid:** cracked, warped, or missing its label well. Replace it before the next episode.
- **Defective tool:** a bent scoop, or tongs that do not close on a piece or do not spring open under a correct tongs grip. Replace it before the next episode.
- **Component out of spec:** a bin 1 component sticky enough that a level scoop will not hold its edge, a bin 2 piece too large to sit inside the left compartment, or bin 3 pieces stuck together so one pick lifts two. Replace the bin contents before the next episode.
- **Container slides** under a correct container hold because the table is wet or slick. Wipe and dry the table before the next episode.

Annotation subtasks (from SOP)

1. Move a container from the container row to the fill spot and take the container hold
2. Portion two level scoops of component 1 into the large compartment
3. Portion four pieces of component 2 into the left compartment
4. Portion three pieces of component 3 into the right compartment

5. Seat the lid on the container
6. Carry the lidded container to the stack spot and square it on the stack
7. Repeat subtasks 1 to 6 for the second, third, and fourth containers
8. Lay the label tile in the label well of the top lid
9. Confirm the stack and end the episode